SECURITY LOG FUNDAMENTALS

Practical Task Report

Objective: To study sample security logs and identify useful fields such as timestamp, user, source IP address, and event type, then use these fields to recognise patterns that may need investigation.

Tools used: Windows Notepad and synthetic sample authentication logs.

1. Sample Log Data

The following five entries were created for practice. They are synthetic examples, not logs collected from a real computer.

Timestamp	User	Source IP	Event Type
2026-10-22 09:10:15	rahul	192.168.1.10	LOGIN_SUCCESS
2026-10-22 09:12:30	admin	203.0.113.25	LOGIN_FAILED
2026-10-22 09:12:45	admin	203.0.113.25	LOGIN_FAILED
2026-10-22 09:13:01	admin	203.0.113.25	LOGIN_FAILED
2026-10-22 09:20:10	priya	192.168.1.15	LOGIN_SUCCESS

2. Important Log Fields

Field	Meaning
Timestamp	The date and time when the event happened.
User	The account associated with the event.
Source IP	The IP address recorded as the source of the activity.
Event type	The result or kind of event, such as LOGIN_SUCCESS or LOGIN_FAILED.

3. Analysis and Observations

• Two successful login events were recorded: one for rahul and one for priya.

• Three failed login events were recorded for the admin account.

• All three failed attempts show the same source IP address: 203.0.113.25.

• The failed attempts occurred close together in time, which is a pattern worth checking.

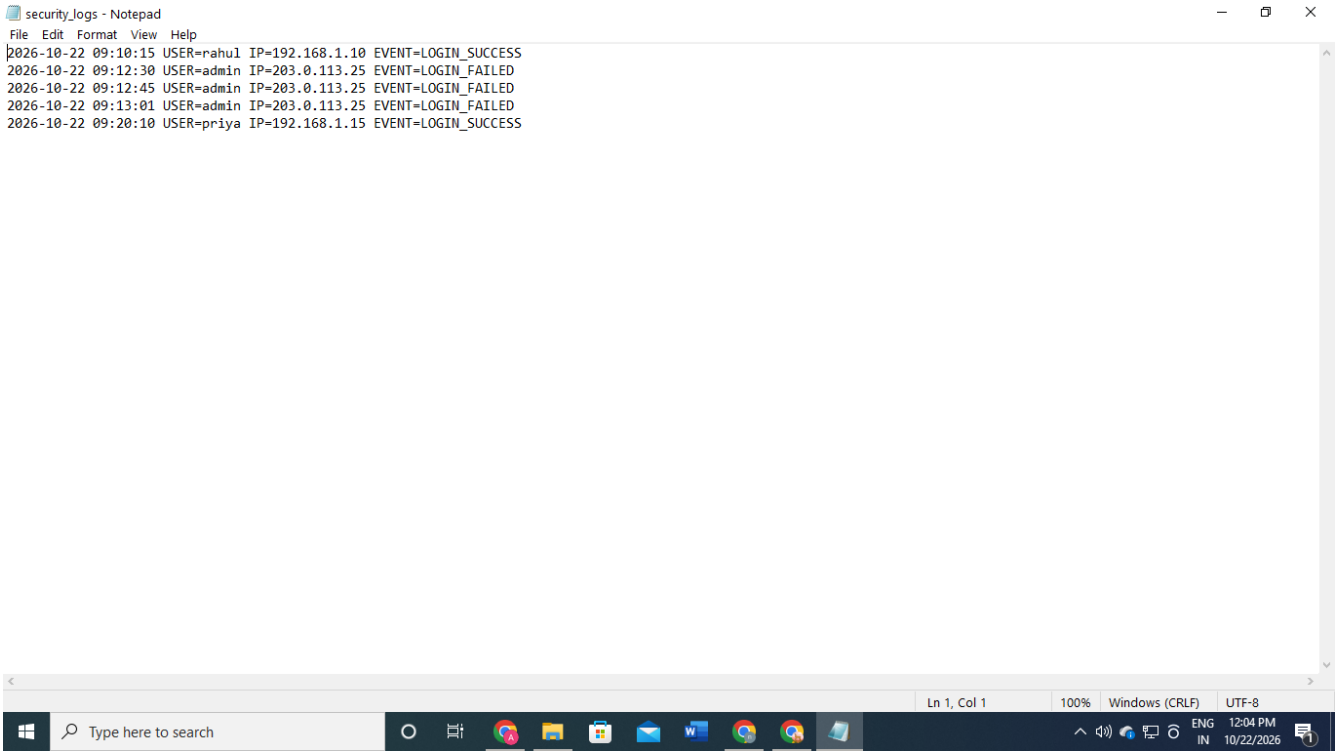
• Repeated failures may indicate password guessing, but these sample entries alone do not prove an attack.

4. Conclusion

This practical demonstrates how to read sample authentication logs, separate successful from failed logins, identify key fields, and notice repeated activity. Security logs help analysts investigate unusual events. In a real investigation, the source, account history, time, and other evidence should be checked before deciding whether an attack occurred.

5. Screenshot Evidence

The screenshot below shows the sample security log file opened in Windows Notepad.



6. Interview Questions and Answers

Question	Answer
What is a security log?	A record of activities or events on a computer or network.
What is a timestamp?	The date and time when an event occurred.
Why are logs important?	They help identify unusual activity, troubleshoot issues, and investigate security events.